SOP-15: Compressed Carbon Dioxide Gas Cylinders

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 213
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	________________________________ Date: _____________
This SOP was created by (if not PI)	________________________________ Date: _____________
Emergency contact	9-1-1 (campus phone) | EH&S 206-543-7262 (M-F 8-5) | 206-685-UWPD (8973) after hours
Date of last review	_______________ Next review due: _______________ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Carbon dioxide, compressed/liquefied gas, 55 lb cylinder	124-38-9

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Gas cylinder (gases under pressure, liquefied gas)

- Stock chemical: one 55 lb cylinder of liquefied carbon dioxide. Intermediates: none - the gas is used as supplied. Uses in this laboratory: bubbling CO2 into seawater to control pH and carbonate chemistry in ocean acidification experiments, and potentially as a euthanasia or anesthetic agent for fish under an IACUC-approved protocol. Wastes: none; the gas is vented or dissolved in experimental seawater.
- The three hazards are ASPHYXIATION, PHYSICAL ENERGY, and CRYOGENIC BURN.
- Asphyxiation: carbon dioxide is heavier than air, colorless and odorless, and displaces oxygen in enclosed and low-lying spaces. It is also physiologically active - unlike a simple asphyxiant, CO2 causes headache, breathlessness, confusion and unconsciousness at concentrations well below oxygen deficiency. A full 55 lb cylinder releasing into a small, poorly ventilated room can produce an immediately dangerous atmosphere. FTR 213 must have adequate general ventilation whenever CO2 is in use, and the room must never be sealed or have its ventilation defeated while the cylinder valve is open.

- Physical energy: a compressed gas cylinder contains enormous stored energy. If a cylinder falls and the valve is sheared off, the cylinder becomes an uncontrolled projectile capable of penetrating walls and causing fatal injury. Every cylinder must be secured upright by a chain or strap at all times, in use and in storage, with the valve protection cap in place whenever a regulator is not attached.
- Cryogenic burn: liquefied CO_2 released rapidly forms dry ice and very cold gas at -78 degC, which causes frostbite on skin contact and can embrittle materials. Rapid discharge can also freeze and block a regulator.
- Routes of exposure: inhalation (the only significant route), and skin contact with cold gas or dry ice.
- How exposure might occur: leaking cylinder or fitting; regulator failure; excessive bubbling into an experimental system in a small room; working in a low-lying or enclosed space where CO_2 has accumulated; and contact with cold discharge.
- Target organs: respiratory system and central nervous system; skin in the case of cryogenic contact.
- Signs and symptoms of exposure: at 2-3% CO_2, increased breathing rate and headache; at 5%, breathlessness, headache, dizziness and sweating; above 8-10%, confusion, loss of consciousness and death. Because CO_2 acts quickly and impairs judgment, a person may not recognize the problem in time to leave. If anyone in the room develops an unexplained headache or breathlessness while CO_2 is in use, everyone leaves the room immediately and the cylinder valve is closed only if that can be done on the way out.
- GHS classification (from MyChem / SDS): Gases under pressure (liquefied gas). MyChem fire class: Inert Compressed Gas.
- Occupational exposure limits: carbon dioxide 5,000 ppm (0.5%) 8-hour TWA and 30,000 ppm (3%) 15-minute STEL (WISHA/OSHA); IDLH is 40,000 ppm (4%). These limits are the basis for requiring ventilation and, where the risk assessment warrants, an oxygen or CO_2 monitor.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- The cylinder is secured upright at all times with a chain or strap at approximately two-thirds of its height, to a wall bracket or a cylinder stand rated for the cylinder size. A cylinder is never left free-standing, leaned against a bench, or secured only by a rope or a bungee cord.
- The valve protection cap is in place whenever a regulator is not attached, including during all transport and storage.
- A CBM-rated (CGA 320) pressure regulator appropriate for carbon dioxide is used. Regulators are not interchanged between gas services, are inspected for damage before each connection, and are replaced rather than repaired in the lab. No thread adapters, thread tape on CGA fittings, or oil or grease on any fitting.
- FTR 213 has general mechanical ventilation. CO_2 is used only with that ventilation operating. Bubbling CO_2 into an open seawater system releases gas into the room; the release rate must be low enough that room ventilation keeps the concentration well below the 0.5% 8-hour TWA.
- For any experiment that will deliver CO_2 continuously, especially unattended or overnight, a continuous CO_2 or oxygen-deficiency monitor with an audible alarm must be installed in the room, positioned at breathing height and near the point of release, and EH&S must be consulted on the installation. An oxygen deficiency hazard assessment is requested from EH&S before any new continuous-delivery setup begins.
- Excess-flow devices or flow restrictors are used where available so that a failed line cannot dump the cylinder contents into the room.
- The nearest eyewash and safety shower are located in FTR 213 and are inspected and flushed weekly by lab staff.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: standard buttoned barrier lab coat.
- Insulated or cryogenic-rated gloves are worn if there is any possibility of contact with cold gas, dry ice, or a rapidly venting line.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: leather or general-purpose work gloves when moving or securing a cylinder, to protect against pinch and crush injury. Insulated or cryogenic-rated gloves when contact with cold discharge or dry ice is possible. Nitrile gloves are worn for the wet-lab portions of seawater work but provide no thermal or mechanical protection.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 safety glasses with side shields whenever a cylinder is being connected, disconnected, moved, or pressurized, and whenever a system under CO2 pressure is being adjusted. A face shield is worn if a line under pressure must be worked on.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP when they are performed as written with adequate ventilation.
- Carbon dioxide may be used outside of a fume hood, in FTR 213, provided the room's general ventilation is operating and the release rate is low. It must NOT be used in a closed room, a walk-in cold room, an environmental chamber, a closet, or any confined or low-lying space.
- An air-purifying respirator DOES NOT protect against carbon dioxide or oxygen deficiency. If an atmosphere cannot be kept safe by ventilation, the answer is a supplied-air respirator or self-contained breathing apparatus and a confined-space or oxygen-deficiency program administered by EH&S - not a cartridge respirator.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.

- No lab member is enrolled in the EH&S Respiratory Protection Program for this work, and none should be without an EH&S oxygen-deficiency hazard assessment first.

Section 4 - Special Handling and Storage Requirements

- Cylinder transport: always move a cylinder with the valve protection cap in place, secured upright on a cylinder cart with a chain or strap. Never roll, drag, slide or carry a cylinder, never lift it by the cap, and never transport it in a passenger elevator with people if it can be avoided - send it up alone and meet it at the destination.
- Connecting a regulator: confirm the cylinder is secured; remove the cap; briefly crack the valve away from people to clear debris (only if the SDS and the gas service permit; for CO_2 this is done with the valve pointed away and no one downstream); inspect the CGA 320 fitting and the regulator threads and the washer; hand-start the nut and tighten with the correct wrench; close the regulator adjusting knob before opening the cylinder valve; open the cylinder valve slowly, standing to the side of the regulator rather than in front of it.
- Leak check every connection with a soap solution or an electronic leak detector after connecting, and again after any change. Never use a flame to check for a leak.
- Cylinder valve is closed and the line depressurized whenever the cylinder is not actively in use, including overnight, unless a continuous-delivery experiment with monitoring is running under an approved setup.
- Never allow a cylinder to be emptied completely; leave a positive residual pressure (approximately 25 psi) so that contaminants cannot back-flow into the cylinder, and mark the cylinder 'MT' when it is spent.
- Every cylinder is labeled with its contents; a cylinder whose label is missing or illegible is never used and is returned to the supplier as unknown. Color alone is not identification.
- Cylinders are dated on receipt and, if not in use, returned to the supplier rather than stored indefinitely; long-held cylinders accrue rental charges and become inventory hazards.
- Storage: secured upright in the designated cylinder location in FTR 213, in a well-ventilated area, away from heat, ignition sources, electrical circuits and areas of heavy traffic or exit routes. Full and empty cylinders are segregated and labeled. Cylinders are not stored in a corridor, stairwell, or means of egress.
- Unattended operations: continuous CO_2 delivery to an experimental system may run unattended only with an installed and functioning CO_2 or oxygen monitor with audible alarm, a posted sign at the door giving the chemical, the responsible person, a contact number and the alarm meaning, EH&S consultation on the setup, and a flow restrictor limiting the maximum possible release.
- A 'COMPRESSED GAS - CARBON DIOXIDE IN USE - ASPHYXIATION HAZARD' sign is posted at the room entrance whenever CO_2 is being delivered to an open system.
- Cleaning: no chemical cleaning is required. Wipe the regulator and fittings free of dust; never apply oil, grease or solvent to any cylinder fitting.
- When work is completed, close the cylinder valve, vent and disconnect the regulator if the cylinder is being taken out of service, replace the valve protection cap, confirm the cylinder is secured, remove gloves and wash hands.

Section 5 - Spill and Accident Procedures

Spill response

- A compressed gas incident is a release, not a spill; the response is evacuation and ventilation, not cleanup.
- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.

- SMALL LEAK detected by soap solution at a fitting, with the cylinder valve accessible and the room atmosphere clearly normal: close the cylinder valve, back off the regulator, and re-make the connection. If the leak persists, close the valve, tag the cylinder 'DEFECTIVE - DO NOT USE', move it to a well-ventilated area if it is safe to do so, and contact the gas supplier and EH&S.
- LEAK AT THE CYLINDER VALVE that cannot be stopped, or ANY release that produces symptoms in anyone present (headache, breathlessness, dizziness, confusion), or any release in an enclosed space: EVACUATE IMMEDIATELY. Close the valve only if it is directly on your way out and can be done in seconds. Close the door behind you, post a warning, prevent entry, and call 9-1-1 from a campus phone. Do not re-enter to retrieve equipment or to investigate. Notify EH&S.
- Carbon dioxide is invisible and odorless and pools at low level. A person who enters to rescue another can be overcome in seconds. Never enter a suspected CO2-enriched space without EH&S or emergency responders and appropriate atmosphere-supplying respiratory protection.
- DRY ICE formation from a rapid release: do not touch it. Allow it to sublime in a ventilated space; do not place it in a sealed container, which will rupture.
- CYLINDER FALLS or the valve is damaged: evacuate immediately and call 9-1-1. Do not attempt to right or restrain a cylinder that is venting.
- PPE for a leak check or a re-made fitting: safety glasses, work gloves, lab coat, full-length pants and close-toe shoes. There is no PPE that makes entering a CO2-enriched atmosphere safe.
- Signage and entry restrictions: post 'DO NOT ENTER - ASPHYXIATION HAZARD - CO2 RELEASE' on the door; restrict entry until EH&S has measured the atmosphere and cleared the space.
- Area cleanup: after EH&S clears the space, no chemical decontamination is required. Ventilate thoroughly and verify the atmosphere before reoccupying.
- There are no spill cleanup materials to dispose of for a gas release. A defective cylinder is returned to the supplier; it is not disposed of as laboratory waste.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately, but only after the rescuer is safe. Do not enter a CO2-enriched space to reach a casualty.
- Inhalation exposure / person collapsed or confused in a CO2 area: do NOT enter. Call 9-1-1 immediately and report a possible asphyxiation and carbon dioxide release. If the person can be reached safely from outside the affected zone, move them to fresh air. Administer oxygen or rescue breathing only if trained and only in a safe atmosphere.
- Anyone who has been in a CO2-enriched atmosphere, even briefly, and who has a headache, breathlessness, dizziness or confusion must be moved to fresh air and evaluated medically; symptoms can persist and judgment may be impaired.
- Cryogenic burn or frostbite from cold gas or dry ice: move to a warm area, remove any clothing that is not frozen to the skin, and warm the affected area with lukewarm (not hot) water at 40-42 degC for 15-20 minutes. Do not rub the area, do not use dry heat, and seek medical evaluation.
- Eye exposure to cold gas: use the eyewash with tepid water for 15 minutes while holding the eyelids open and seek medical evaluation.
- Skin exposure other than cold: not applicable; carbon dioxide gas is not a skin irritant.

- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (carbon dioxide), estimated concentration or duration, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and post the hazard.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

Section 6 - Waste Accumulation and Disposal Procedures

- No chemical waste is generated by the use of compressed carbon dioxide. The gas is either dissolved in experimental seawater or vented to the room ventilation.
- EMPTY CYLINDERS ARE NOT WASTE and must never be placed in the trash, in a dumpster, in scrap metal recycling, or in a hazardous waste pickup. They are the property of the gas supplier and are returned to the supplier for refilling.
- Mark a spent cylinder 'MT' or 'EMPTY', close the valve, replace the valve protection cap, secure it upright in the designated empty-cylinder location segregated from full cylinders, and arrange return with the supplier promptly.
- A cylinder that is defective, leaking, of unknown contents, has an illegible label, or is past its hydrostatic test date is tagged 'DEFECTIVE - DO NOT USE', secured in a ventilated area, and reported to both the supplier and EH&S. Do not attempt to vent, empty or dispose of such a cylinder in the laboratory.
- Experimental seawater acidified with CO_2 contains no added hazardous constituent from this SOP and may be handled per the experimental protocol and the applicable seawater discharge permit; if it contains any chemical from another SOP (for example copper under SOP-12), that SOP's waste requirements govern.
- Dry ice formed accidentally is allowed to sublime in a ventilated area; it is never placed in a sealed container or a drain.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Update the MyChem inventory when a cylinder is received, returned or exchanged.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials. Update the MyChem inventory when containers are emptied, received, transferred or surplused.

Section 7 - Protocol

- Procedure A - Receiving and securing a cylinder. Confirm the label matches what was ordered and is legible. Move the cylinder on a cylinder cart, capped and strapped. Secure it upright in the designated location with a chain or strap at two-thirds height before removing it from the cart. Record receipt in MyChem.
- Procedure B - Connecting the regulator. With the cylinder secured, remove the cap, inspect the CGA 320 outlet and the regulator inlet and washer for damage, dirt, oil or grease. Hand-start and then wrench-tighten the nut. Close the regulator adjusting knob fully. Stand to the side and open the cylinder valve slowly until fully open. Set the delivery pressure. Leak-check every joint with soap solution.
- Procedure C - Delivering CO_2 to a seawater system. Confirm room ventilation is operating and, for continuous delivery, that the CO_2/oxygen monitor is powered and in alarm-ready state. Set the lowest flow that achieves the

target pH. Post the room entrance sign. Log the start time, flow setting and responsible person. Check the system and the monitor at the interval specified in the experimental protocol.

- Procedure D - Shutting down. Close the cylinder valve. Allow the line to depressurize through the system or a vent, then back off the regulator adjusting knob. If the cylinder is being taken out of service, disconnect the regulator and replace the valve protection cap. Confirm the cylinder remains secured.
- Procedure E - Returning an empty cylinder. Leave approximately 25 psi residual, close the valve, mark 'MT', cap it, secure it in the empty-cylinder location, and arrange supplier pickup.
- Prohibited: using CO2 in a closed room, cold room or environmental chamber; leaving a cylinder unsecured; transporting a cylinder without a cap and cart; using oil, grease, thread tape or adapters on fittings; using a flame to leak-check; entering a suspected CO2-enriched space; disposing of a cylinder as waste; unattended continuous delivery without a monitor.
- Refer to the UW EH&S Compressed Gases and Cryogens guidance and to Section 2 of the UW Laboratory Safety Manual for additional requirements.
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

Potentially yes. If carbon dioxide is used for euthanasia or anesthesia of fish or other animals, that use must be documented and approved by IACUC, must follow the current AVMA Guidelines for the Euthanasia of Animals, and must use a dedicated, properly regulated delivery setup with a controlled displacement rate. Personnel performing CO2 euthanasia protect themselves by working in a well-ventilated room with the euthanasia chamber closed during delivery, never leaning over an open chamber, verifying room ventilation before starting, and evacuating the chamber's atmosphere to the room ventilation rather than into the breathing zone. Animal carcasses and holding water from CO2 euthanasia carry no chemical residue and are handled per the IACUC protocol. If CO2 is used to acidify seawater holding live animals, that exposure is conducted under the applicable IACUC-approved protocol.

IACUC protocol number (if applicable): ______________________________

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 9 - Approvals Required

- All staff working with compressed carbon dioxide must be trained on this SOP prior to starting work. They must also review the carbon dioxide SDS, and it must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) documented review of this SOP, the SDS, and the UW EH&S Compressed Gases and Cryogens guidance; (3) documented hands-on instruction on cylinder securing, transport, regulator connection and leak checking.
- A worker must complete at least one supervised cylinder change and regulator connection (Procedures A, B and D), observed and signed off by the PI or a designated experienced lab member, before doing it independently.
- Any continuous or unattended CO2 delivery setup requires prior written PI approval AND consultation with EH&S for an oxygen deficiency hazard assessment and monitor specification before the experiment begins.
- Use of CO2 for animal euthanasia or anesthesia requires a current IACUC-approved protocol and documented training in that procedure.

- No medical examination is required for the work described because respirators are not used. Any work requiring supplied-air respiratory protection would require medical evaluation, training and fit testing through the EH&S Respiratory Protection Program, and an EH&S oxygen deficiency assessment first.

Section 10 - Decontamination

- No chemical decontamination is required for carbon dioxide; the gas leaves no residue.
- Equipment: wipe the regulator body, gauges and fittings with a clean dry cloth to remove dust. NEVER apply oil, grease, solvent or thread tape to any cylinder or regulator fitting - contamination of a fitting is a safety defect.
- Delivery tubing and diffusers used in seawater are rinsed with fresh water and air-dried; if they were used in a system containing another chemical, that chemical's SOP governs their decontamination.
- Cylinder exterior: wipe clean before returning to the supplier; ensure the label is intact and the 'MT' marking is visible.
- Controlled areas: there are no glove boxes or restricted-access hoods involved. The designated cylinder location in Section 11 is kept clear, dry and free of stored combustibles.
- Cleaning solutions and materials: clean dry cloths, fresh water for tubing, and soap solution for leak checking only.

Section 11 - Designated Area

- No designated area is required because carbon dioxide is not a particularly hazardous substance. Sections 9-11 are provided for completeness and are optional for this SOP; they are included because the asphyxiation hazard warrants the same discipline.
- Cylinder location: the designated, chained cylinder position in FTR 213, in a well-ventilated area away from heat, ignition sources, electrical panels, exits and traffic routes. Full and empty cylinders are segregated and labeled at this location.
- Use location: FTR 213, with general ventilation operating, at the seawater experimental system. The room entrance is posted 'COMPRESSED GAS - CARBON DIOXIDE IN USE - ASPHYXIATION HAZARD' during delivery to an open system.
- Prohibited locations: any cold room, environmental chamber, closet, corridor, stairwell or other confined, unventilated or egress space. CO_2 is never delivered or stored in these.
- Where a continuous-delivery experiment is running, the monitor location and alarm set point are recorded here and the room is posted with the meaning of the alarm and the required response.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.

I have read and understand the content of this SOP:

Name (print)	Signature	Date